

Non-Clinical Interventions to reduce overall cesarean sections

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Executive summary -

Introduction →

The number of women who deliver by Cesarean section has increased steadily in Egypt over the last three decades, from 4.6% in 1992 (1) to 51.8% in 2014 (2)

Cesarean birth is associated with short- and long-term risks that can extend many years beyond the current delivery and affect the health of the woman, the child, and future pregnancies. High rates of cesarean section are associated with substantial health-care costs (3)

The factors contributing to the rise in cesarean section rates are complex, and identifying interventions to address them is challenging. Factors associated with cesarean births include changes in the characteristics of the population such as increase in the prevalence of multiple pregnancies, assisted reproduction, and increase in the proportion of nulliparous women with advanced age. Other non-clinical factors such as women increasingly wanting to determine how and when their child is born, generational shifts in work and family responsibilities, physician factors, increasing fear of medical litigation, as well as organizational, economic and social factors have all been implicated in this increase (4)

The sustained, unprecedented rise in cesarean section rates in Egypt is a major public health concern. There is an urgent need for evidence-based guidance to address the trend. This is the national guideline on non-clinical interventions (defined as interventions applied independently of a clinical encounter between a health-care provider and a patient in the context of patient care)

The objective of this guideline is to provide evidence-based recommendations on non-clinical interventions specifically designed to reduce cesarean sections. Clinical interventions that could help to reduce cesarean section rates are being prepared in a second guideline

Target audience →

The primary audience for this guideline includes health-care professionals including obstetricians, midwives, nurses, and general medical practitioners, as well as managers of maternal and child health programs and public health policymakers in Egypt

Guideline development methods →

We used the GRADE-Adolopment methodology for the guideline adaptation process (5). Briefly, this included, identifying and training guideline panelists, prioritizing questions and outcomes, identifying existing guideline (6), assessing quality and adaptability of the identified guideline, reviewing GRADE evidence tables and EtD frameworks, and formulating and grading strength of recommendations



The identified existing guideline was developed by the WHO in accordance with standard procedures set out in the WHO *handbook for guideline development* (7). The evidence, in the identified guideline, on the effectiveness of interventions was derived from an updated Cochrane .review of 29 studies (8)



Recommendations -

The recommendations are intended to inform national and subnational policies and protocols to reduce cesarean births. They should be implemented alongside the clinical recommendations for intrapartum care to improve the quality of care for mothers and newborns during childbirth. The :recommendations are

[Recommendation 1](#)

Implementation of evidence-based clinical practice guidelines (on-site training, facilitation by a local opinion leader [obstetrician-gynaecologist], supervision) plus audits of indications for cesarean delivery and provision of feedback to healthcare professionals are recommended to reduce overall .cesarean sections

.(Recommended, High-certainty evidence)

[Recommendation 2](#)

Healthcare professionals' education by a local opinion leader [obstetrician-gynaecologist] to promote implementation of evidence-based clinical practice guidelines is recommended to reduce .repeat cesarean sections

.(Recommended, High-certainty evidence)

[Recommendation 3](#)

Implementation of evidence-based clinical practice guidelines combined with structured, mandatory second opinion for cesarean section indication is recommended to reduce cesarean births only with targeted monitoring and evaluation in settings with an on-site senior clinician able to provide mandatory second opinion for cesarean section indication. *(Context-specific .recommendation, moderate-certainty evidence)*

[Recommendation 4](#)

For the sole purpose of reducing cesarean sections, peer review plus mandatory second opinion .are recommended only in the context of rigorous research

.(Context-specific recommendation, very low-certainty evidence)

[Recommendation 5](#)

For the sole purpose of reducing cesarean sections, audit and feedback plus financial incentive for health-care professionals or health-care organizations are recommended only in the context of .rigorous research

.(Context-specific recommendation, very low-certainty evidence)



[Recommendation 6](#)

For the sole purpose of reducing cesarean sections, audit and feedback using Robson classification
.is recommended only in the context of rigorous research
.(Context-specific recommendation, very low-certainty evidence)



Acknowledgements -

The Egyptian Guideline Development Group gratefully acknowledge the contributions that many individuals and organizations have made to the development of this guideline

The members of the Steering Group who managed the guideline development process were: Mohamed Hamed Salama, Hamdy Bakry Alqenawy, Asmaa Fahmy Kassem, Yasmeen Nashaat, Essam Eldin Abdelrehim, and Ashraf Fawzy Nabhan. Ashraf Nabhan coordinated the guideline development project and chaired the GDG

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Acronyms and abbreviations

Developing and Evaluating Communication strategies to support Informed Decisions and practice based on Evidence	DECIDE
Evidence to Decision	EtD
Guideline Development Group	GDG
GRADE Evidence to Decision frameworks for adoption, adaptation, and de novo development of trustworthy recommendations	GRADE-ADOLPMENT
Grading of Recommendations Assessment, Development and Evaluation	GRADE
Guideline-driven, Research priorities, Evidence synthesis, Application of evidence and Transfer of knowledge	GREAT
Nulliparous women with a term, singleton baby in a vertex position	NTSV
Population, Intervention, Comparator, Outcome	PICO
World Health Organization	WHO

General scope of the guideline

Non-clinical interventions to reduce cesarean sections	Title
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to reduce cesarean sections	Purpose
an individual patient, community	Perspective
low-risk pregnancy	Target population
prior cesarean section should be considered when making recommendations	Key coexisting conditions
health care level providing labor and delivery services	Setting
Preventive	Types of interventions
obstetricians, nurses, general medical practitioners, managers of maternal and child health programs and public health policy-makers	Key stakeholders and users
need for adequate human resources, appropriate infrastructure, and supportive system changes (financial, legislative)	Key resources to consider
Several barriers may hinder the effective implementation and scale-up of the recommendations in this guideline. These factors include behaviors of patients (women or families), the behavior of health-care professionals, to the organization of care, health service delivery or to .financial arrangements	Key issues for implementation
WHO recommendations non-clinical interventions to reduce unnecessary cesarean sections. Geneva: World Health Organization; .2018	Existing guideline that has been adapted



Introduction -

[Background](#)

Significant improvements in clinical obstetric care and increased safety in surgical procedures have been observed over the past three decades. This appears an increase in cesarean sections in Egypt, like a global trend. The number of women who deliver by Cesarean section has increased steadily .in Egypt over the last three decades, from 4.6% in 1992 (1) to 51.8% in 2014 (2)

Cesarean birth is associated with short- and long-term risks that can extend many years beyond the current delivery and affect the health of the woman, the child, and future pregnancies. As with any surgery, cesarean section is associated with short- and long-term risks. These can extend many years beyond the current delivery and affect the health of the woman, the child, and future pregnancies. Cesarean section increases the likelihood of requiring a blood transfusion, the risks of anesthesia complications, organ injury, infection, thromboembolic disease, and neonatal respiratory distress, among other short-term complications. Cesarean section has been associated in the long term with an increased risk of asthma and obesity in children, and complications in subsequent pregnancies, such as uterine rupture, placenta accreta, placenta previa, hysterectomy, and intra-abdominal adhesions, with the risk of these morbidities progressively increasing as the number of previous cesarean deliveries increases. These risks are higher in women with limited access to comprehensive obstetric care, and they require careful consideration in settings that lack the facilities and capacity to conduct surgery safely or to treat surgical complications. High rates of .cesarean section are associated with substantial health-care costs (3)

The factors contributing to the rise in cesarean section rates are complex, and identifying interventions to address them is challenging. Factors associated with cesarean births include changes in the characteristics of the population such as increase in the prevalence of multiple pregnancies, assisted reproduction, and increase in the proportion of nulliparous women with advanced age. Other non-clinical factors such as women increasingly wanting to determine how and when their child is born, generational shifts in work and family responsibilities, physician factors, increasing fear of medical litigation, as well as organizational, economic and social factors .have all been implicated in this increase (4)

For nearly 30 years, the international health-care community has considered the ideal rate for cesarean section to be between 10% and 15%. WHO released a statement summarizing the results of systematic reviews and analysis of the available data on cesarean births. Considering the evidence, the panel of experts convened by WHO concluded in the statement that, at population level, cesarean section rates higher than 10% were not associated with reductions in rates of maternal and newborn mortality. The statement notes, however, that the association between cesarean section rates and other relevant outcomes such as stillbirths, maternal and perinatal



morbidity, pediatric outcomes and psychological or social well-being could not be determined due to the lack of data on these outcomes at the population level. The scarcity of data is a limitation of this evidence that needs to be borne in mind when interpreting the WHO statement (9)

The ideal or optimal cesarean rate is unknown, yet cesarean section is effective in saving maternal and infant lives, but only when it is used for medically indicated reasons. Ultimately, every effort should be made to provide cesarean sections to women in need, rather than striving to achieve a specific rate

This is the first national guideline on non-clinical interventions to reduce cesarean sections – i.e. those performed in the absence of medical indications. Non-clinical interventions in this guideline refer to those interventions applied independently of a clinical encounter between a particular provider and patient in the context of patient care

Why this guideline is needed →

The rise in cesarean section rates is a national problem. The causes of the increase are multiple. Changes in the characteristics of the population, such as the increase in assisted reproduction, advanced maternal age, and multifetal gestation, have been cited to contribute to the rise. Other factors such as differences in style of professional practice, increasing fear of medical litigation, and organizational, economic, social, and cultural factors have all been implicated in this rising tide

We expect this guideline to help clinicians and other health-care professionals to reduce rates of cesarean sections. Effective implementation of this guideline will contribute to achievement of the United Nation's Sustainable Development Goal 3 ("Ensure healthy lives and promote well-being for all at all ages") by improving the quality of care during childbirth and reducing complications, disability and death associated with cesarean births, particularly in settings that lack the facilities and/or capacity to properly conduct safe surgery (10)

Target audience →

The primary audience for this guideline includes health-care professionals namely obstetricians, midwives, nurses, and general medical practitioners, as well as managers of maternal and child health programs and public health policymakers in Egypt



Methods -

The GDG followed the GRADE-Adolopment methodology to develop this guideline (6)

Topic prioritization and need for a guideline: has been decided by the Supreme Health Council (1)
.based on the importance of the topic and the potential impact of the guideline when implemented

The intention of the guideline is to improve practice and to change health policies. The target (2)
.audience and scope of the document were explicitly defined at the outset

.Setting up the guideline development group. All members signed a COI statement (3)

Identification and assessment of existing guidelines: The GDG screened potentially relevant (4)
guidelines. The development group assessed the relevance, timeliness, and quality of these
.reports. The GDG identified only one comprehensive, high-quality, and up-to-date guideline (6)

The identified guideline Assessment of the methodological quality of the guideline using the (5)
:AGREE GRS instrument (11). AGREE GRS includes

i) Overall quality of guideline development methods

ii) Overall quality of guideline presentation

iii) Completeness of reporting

iv) Overall quality of guideline recommendations

v) Overall assessment of the guideline

b) Adaptability assessment: The GDG assessed the adaptability of the original document. Among
the adaptability assessment, the GDG focused on the evidence profiles, the evidence to decision
.tables, and the inclusion of the pre-specified critical outcomes

c) Decision: After the guideline assessment, the GDG decided to adapt the original document,
.taking into consideration the resources required for the process

d) Evidence profiles

i) The GRADE approach (GRADEpro GDT) was used to create the evidence profiles. This
process involved two main steps: (1) the evaluation of the certainty of the evidence and (2) the
.summary of findings tables

ii) For guideline panels, the certainty of evidence reflects “the extent to which the confidence
in an estimate of the effect is adequate to support a particular recommendation”. When
assessing this item, the GDG evaluated the study design, the consistency of the results across



the studies, the precision of the results, the directness of the evidence, the likelihood of publication bias, and the magnitude of the effect. The certainty of evidence is categorized as .high, moderate, low, or very low

iii) The second step consists of the summary of findings table. This chart shows the results of the studies, using both relative and absolute measures, indicating the total number of patients in each group, the total number of events, an estimate of the control group risk, the effect size, .and the certainty of evidence for each outcome

e) Evidence to decision tables: The chair of the GDG presented the evidence profiles and the EtD tables in the face-to-face meeting and the group discussed the final recommendations. The GDG determined the direction and strength of the final recommendations using the GRADE methodology, based on four key factors presented in the evidence profiles and EtD tables: the balance between benefits and harms, certainty of evidence, patients' preferences and values, and resources considerations (including equity and feasibility). The panel decided the final redaction of .the recommendations

.f) Final draft: The GDG collaborated in writing the final draft of the clinical guideline

g) External revision: The guideline underwent peer review before final publication. Experts that are .not in the development group performed the external revision



Priority questions and outcomes -

:The population of interest comprised

women seeking antenatal, labour and delivery care in health-care facilities (term, singleton, in
;vertex presentation with or without a prior cesarean)

;families of pregnant women

health-care professionals who work with expectant mothers (midwives, nurses and obstetricians);
and

health-care facilities that provide maternity care; and

:Interventions

Opinion leaders to improve adherence to evidence based clinical guidelines .1

Education of healthcare professionals to improve adherence to evidence-based clinical .2
guidelines

Mandatory second opinion for cesarean indication .3

Audit and feedback .4

Peer review .5

Continuous support in labour .6

hour in-house obstetrician coverage-24 .7

Midwife-led continuity-of-care model .8

Health education of women in pregnancy .9

Education of the community .10

Goal setting for cesarean rates .11

Public dissemination of cesarean rates .12

Hospital quality improvement programme .13

.Targeted financial strategies for health-care professionals or health-care organizations .14

Policies that limit financial/legal liability in case of litigation of healthcare professionals or .15
.organizations

Comparator: usual care



:Critical outcome

.Overall cesarean sections .1

.Repeat cesarean sections .2

.NTSV Cesarean sections .3

Related guidelines

This guideline is a companion to the guideline intended to provide recommendations on the best
.intrapartum care

Focus and approach

.The focus of this guideline is non-clinical interventions for reducing cesareans



Formulation of the recommendations -

The DECIDE framework was used to guide the formulation of recommendations. DECIDE is an evidence-to-decision tool that allows explicit and systematic consideration of evidence on interventions in terms of six domains: effects, values, resources, equity, acceptability and feasibility. For each priority question, judgements are made on the impact of the intervention against each of these domains, to inform guideline recommendations

Using the DECIDE framework, members of the GDG prepared evidence-to-decision tables for each priority question, covering evidence on each of these six domains. At the face-to-face meeting (held in July 2023), GDG members collectively reviewed the evidence-to-decision tables and the draft recommendations, and reached consensus on each recommendation, based on explicit consideration of domains within the evidence-to-decision tables. The GDG also identified important considerations for guideline implementation, monitoring and evaluation, and research gaps

The GDG made three types of recommendation

Recommended: The benefits of implementing this option outweigh the possible harms. This .1 option can be implemented

Context-specific recommendation .2

Recommended only in the context of rigorous research: This option indicates that there .2.1 are important uncertainties about an intervention. In such instances, the implementation can still be undertaken at a large scale, but only as research that is able to address unanswered questions and uncertainties related both to the effectiveness of an intervention

Recommended only with targeted monitoring and evaluation: This option indicates .2.2 uncertainty about the effectiveness or acceptability of an intervention, especially regarding particular contexts or conditions. Interventions classified as such can be considered for implementation (including at large scale), provided they are accompanied by targeted monitoring and evaluation

No recommendation: The reasons for deciding “No recommendation” are either .3

The confidence in effect estimates is low or very low that the GDG considers a .3.1 recommendation would be speculative

Irrespective of the confidence in effect estimates, the trade-offs are so closely balanced, .3.2 and the values and preferences and resource implications not known or too variable, that the panel has great difficulty deciding on the direction of a recommendation

Decision-making during the GDG meeting →



The GDG meeting was held in July 2023. The GDG members discussed the evidence summarized in the evidence-to-decision tables for each guideline question and then considered the relevant draft recommendation. After discussing each question, the draft recommendation and justification were .revised as needed. The final set of recommendations was made by consensus

Document preparation and peer review ➔

Following the final GDG meeting, a draft of the full guideline document was prepared. The document was sent to the External Review Group for peer review. The External Review Group members were asked to review the final draft guideline to identify errors of fact, comment on clarity of language, and consider issues of implementation, adaptation and context. The Steering Group evaluated the input of the peer reviewers for inclusion in the guideline document and made further revisions to the guideline draft as needed. After the GDG meetings and external peer review, further modifications to the guideline by the Steering Group were limited to corrections of factual error and improvements in language to address any lack of clarity. The revised final version .was returned to the GDG for the final approval



Evidence and recommendations -

Recommendation 1

Implementation of evidence-based clinical practice guidelines (on-site training, facilitation by a local opinion leader [obstetrician-gynaecologist], supervision) plus audits of indications for cesarean delivery and provision of feedback to healthcare professionals are recommended to reduce overall cesarean sections.

.(Recommended, High-certainty evidence)

Evidence-base

Effects of interventions ■

One cluster RCT (12) assessed the effectiveness of standard of care versus multifaceted intervention comprising implementation of evidence-based clinical practice guidelines (on-site training, facilitation by a local opinion leader [obstetrician-gynecologist], supervision), audits of indications for caesarean delivery and provision of feedback to HCP. High certainty evidence indicates that the multi-faceted intervention reduced overall cesarean sections (RR 0.929 [95% CI 0.900 to 0.959])

Values ■

No direct local research evidence was identified on the values and preferences of women on the implementation of evidence-based clinical practice guidelines, audits and, feedback

Resources ■

No direct local research evidence was identified on the impact on resource use (costs) of implementation of evidence-based clinical practice guidelines, audits and, feedback

Cost-effectiveness ■

No local research evidence was identified on the cost-effectiveness of implementation of evidence-based clinical practice guidelines, audits and, feedback

Equity ■

No direct local evidence was identified on the impact on equity of implementation of evidence-based clinical practice guidelines, audits and, feedback

Acceptability ■

No local research evidence was identified on the acceptability of implementation of evidence-based clinical practice guidelines, audits and, feedback



Feasibility ■

No local research evidence was identified on the Feasibility of implementation of evidence-based clinical practice guidelines, audits and, feedback 

Recommendation 2

Healthcare professionals' education by a local opinion leader [obstetrician-gynaecologist] to promote implementation of evidence-based clinical practice guidelines is recommended to reduce repeat cesarean sections
(Recommended, High-certainty evidence)

Evidence-base

Effects of interventions ■

One cluster RCT (13) assessed the effectiveness of standard of care versus HCP education by a local opinion leader to promote compliance with guidelines that recommended clinical action to increase trial of labour and vaginal birth rates in women who have had previous caesarean sections. There was high certainty evidence that the multi-faceted intervention reduced repeat cesarean sections (RR 0.879 [95% CI 0.838 to 0.921]) 

Values ■

No direct local research evidence was identified on the values and preferences of women on education by a local opinion leader to promote compliance with guidelines that recommended clinical action to increase trial of labour and vaginal birth rates in women who have had previous caesarean sections 

Resources ■

No direct local research evidence was identified on the impact on resource use (costs) of education by a local opinion leader to promote compliance with guidelines that recommended clinical action to increase trial of labour and vaginal birth rates in women who have had previous caesarean sections 

Cost-effectiveness ■

No local research evidence was identified on the cost-effectiveness of education by a local opinion leader to promote compliance with guidelines that recommended clinical action to increase trial of labour and vaginal birth rates in women who have had previous caesarean sections 

Equity ■



No direct local evidence was identified on the impact on equity of education by a local opinion leader to promote compliance with guidelines that recommended clinical action to increase trial of .labour and vaginal birth rates in women who have had previous caesarean sections

Acceptability ■

No local research evidence was identified on the acceptability of education by a local opinion leader to promote compliance with guidelines that recommended clinical action to increase trial of .labour and vaginal birth rates in women who have had previous caesarean sections

Feasibility ■

No local research evidence was identified on the feasibility of education by a local opinion leader to promote compliance with guidelines that recommended clinical action to increase trial of .labour and vaginal birth rates in women who have had previous caesarean sections

Recommendation 3

Implementation of evidence-based clinical practice guidelines combined with structured, mandatory second opinion for cesarean section indication is recommended to reduce cesarean births **only with targeted monitoring and evaluation** in settings with an on-site senior clinician able to provide mandatory second opinion for cesarean section indication. (*Context-specific .recommendation, moderate-certainty evidence*)

Evidence-base

Effects of interventions ■

One cluster RCT (14) assessed the effectiveness of standard of care versus Implementation of clinical practice guidelines combined with mandatory second opinion for cesarean section indication. **Moderate certainty evidence suggests that Implementation of clinical practice guidelines combined with mandatory second opinion probably reduce overall cesarean sections (RR 0.992 .[95% CI 0.968 to 1.017])**

Values ■

No direct local research evidence was identified on the values and preferences of women on the implementation of evidence-based clinical practice guidelines, combined with structured, .mandatory second opinion

Resources ■

No direct local research evidence was identified on the impact on resource use (costs) of implementation of evidence-based clinical practice guidelines, combined with structured, .mandatory second opinion

Cost-effectiveness ■



No local research evidence was identified on the cost-effectiveness of implementation of evidence-based clinical practice guidelines, combined with structured, mandatory second opinion

Equity ■

No direct local evidence was identified on the impact on equity of implementation of evidence-based clinical practice guidelines, combined with structured, mandatory second opinion

Acceptability ■

No local research evidence was identified on the acceptability of implementation of evidence-based clinical practice guidelines, combined with structured, mandatory second opinion

Feasibility ■

No local research evidence was identified on the feasibility of implementation of evidence-based clinical practice guidelines, combined with structured, mandatory second opinion

Recommendation 4

For the sole purpose of reducing cesarean sections, peer review plus mandatory second opinion are recommended only in the context of rigorous research

.(Context-specific recommendation, very low-certainty evidence)

Evidence-base

Effects of interventions ■

One before-after study (15) assessed the effectiveness of standard of care versus peer review plus mandatory second opinion for reducing the overall cesarean sections. The effect of intervention is uncertain because of very low certainty evidence (RR 0.830 [95% CI 0.796 to 0.866])

Values ■

No direct local research evidence was identified on the values and preferences of women on peer review plus mandatory second opinion

Resources ■

No direct local research evidence was identified on the impact on resource use (costs) of peer review plus mandatory second opinion

v Cost-effectiveness

No local research evidence was identified on the cost-effectiveness of peer review plus mandatory second opinion

Equity ■



No direct local evidence was identified on the impact on equity of peer review plus mandatory 
.second opinion

Acceptability ■

No local research evidence was identified on the acceptability of peer review plus mandatory 
.second opinion

Feasibility ■

No local research evidence was identified on the feasibility of peer review plus mandatory 
.second opinion

Recommendation 5

For the sole purpose of reducing cesarean sections, audit and feedback plus financial incentive for health-care professionals or health-care organizations are recommended only in the context of
.rigorous research

.(Context-specific recommendation, very low-certainty evidence)

Evidence-base

Effects of interventions ■

One before-after study (16) assessed the effectiveness of standard of care versus audit and 
feedback plus financial incentive for reducing the overall cesarean sections. The effect of
.intervention is uncertain because of very low certainty evidence (RR 0.828 [95% CI 0.758 to 0.904])

Values ■

No direct local research evidence was identified on the values and preferences of women on 
.the audit and feedback plus financial incentive

Resources ■

No direct local research evidence was identified on the impact on resource use (costs) of audit 
.and feedback plus financial incentive

Cost-effectiveness ■

No local research evidence was identified on the cost-effectiveness of audit and feedback plus 
.financial incentive

Equity ■

No direct local evidence was identified on the impact on equity of audit and feedback plus 
.financial incentive

Acceptability ■



No local research evidence was identified on the acceptability of audit and feedback plus  .financial incentive

Feasibility ■

No local research evidence was identified on the feasibility of audit and feedback plus financial  .incentive

Recommendation 6

For the sole purpose of reducing cesarean sections, audit and feedback using Robson classification
 .is recommended only in the context of rigorous research
 .(Context-specific recommendation, very low-certainty evidence)

Evidence-base

Effects of interventions ■

One before-after study (17) assessed the effectiveness of standard of care versus audit and  feedback using Robson classification for reducing the overall cesarean sections. The effect of .intervention is uncertain because of very low certainty evidence (RR 0.862 [95% CI 0.765 to 0.972])

Values ■

No direct local research evidence was identified on the values and preferences of women on  .audit and feedback using Robson classification

Resources ■

No direct local research evidence was identified on the impact on resource use (costs) of audit  .and feedback using Robson classification

Cost-effectiveness ■

No local research evidence was identified on the cost-effectiveness of audit and feedback  . using Robson classification

Equity ■

No direct local evidence was identified on the impact on equity of audit and feedback using  .Robson classification

Acceptability ■

No local research evidence was identified on the acceptability of audit and feedback using  .Robson classification

Feasibility ■



No local research evidence was identified on the feasibility of audit and feedback using 
Robson classification



Implementation of the recommendations -

:The necessary steps for implementing a guideline are

- .Analyze local needs and priorities (1)
- .Identify all potential barriers and facilitating factors (2)
- .Determine available resources (3)
- Design a strategy to support the adoption of the recommendations and to make the overall (4)
- .context favorable to the proposed changes

Several barriers may constitute a bottleneck for the effective implementation and scale-up of the recommendations in this guideline. These factors may be related to the behaviors of patients (women or families), the behavior of health-care professionals, to the organization of care, health service delivery or to financial arrangements. The barriers were identified from focus group :discussion. The barriers include

- Maternal request for CS .1
- Financial incentives .2
- Medicolegal threats in the absence of clear and explicit support and protection .3
- Lack of human resources with the necessary expertise and skills to implement, supervise .4
- and support recommended practices (e.g. senior clinicians to provide second opinion for .cesarean section indication)
- Lack of health information management systems designed to document and monitor .5
- .recommended practices (e.g. electronic records, registers)

Monitoring and evaluation

The implementation and the impact of the recommendations will be monitored at the health-service levels based on clearly defined criteria and indicators that are associated with locally agreed targets. The WHO *Standards for improving quality of maternal and newborn care in health facilities* provide lists of prioritized input, output, and outcome measures, which can be used to .define quality-of-care criteria and indicators with locally agreed targets

:Suggested indicators

- Overall Cesarean sections .1
- .a. Denominator: women with a livebirth
- b. Numerator: women in the denominator who had cesarean sections



- a. Denominator: Nulliparous with a livebirth term singleton in vertex presentation
- b. Numerator: women in the denominator who had cesarean sections

Updating the guideline

The recommendations included in this guideline will be regularly reviewed and prioritized as needed by the National Steering Group. The W.H.O recommends a minimum of two years and a maximum of five, but it is essential to take into consideration the pace of change of research on the topic

The National Steering Group will continue to follow the research developments in cesarean section, particularly those relating to questions for which no evidence was found and those that are supported by evidence of very low or low certainty, where new recommendations or a change in the published recommendations may be warranted. Decisions to make updates will also be informed by data on ongoing studies identified from trial registry searches. Following the publication and dissemination of the guideline, any appropriate concern about the validity of any recommendation will be promptly communicated to the guideline implementers in addition to informing plans to update the recommendation

Where there are concerns about the validity of a recommendation based on new evidence, the systematic review addressing the primary question will be updated. Any new questions identified following the scoping exercise at the end of five years will undergo a similar process of evidence retrieval, synthesis and grading in accordance with the standards for guideline development

In conjunction with the Steering Group, there will be periodic assessment of the currency of the recommendations and the need for new or updated guidance on the topic. This will be achieved by performing a scoping exercise among technical experts, health-care professionals, and research and service users to identify controversial or priority areas where evidence-based guidance may be needed

Dissemination

This guideline is available online and as a printed publication. Technical meetings will be held within the national maternity hospitals to share the recommendations and derivative products with the teams responsible for policy and program implementation

The executive summary and recommendations will be translated into Arabic for mainstream media and the public. A policy brief summarizing the recommendations and implementation-related issues will be developed for policymakers and program managers. To increase awareness of the guideline, the recommendations will also be published in a peer-reviewed journal

Research implications



The GDG and Steering Group identified areas where further studies are needed based on four :broad considerations

- .1 uncertainty in the effects of the interventions due to evidence of very low or low certainty
- .2 concerns with the applicability of the evidence (particularly as most interventions were .assessed in single studies; the interventions would benefit from replication)
- .3 .lack of studies for predefined guideline questions
- .4 promising interventions not specifically designed to reduce cesarean births that would benefit from examination in areas with high cesarean section rates (e.g. continuous one-to-one .intrapartum support)

Additional research questions were proposed by the GDG during the face-to-face meeting. In particular, the GDG emphasized that future intervention trials should be preceded with formative research to define locally relevant determinants of cesarean births. Prioritized research gaps :include

Local evidence namely qualitative assessment of patient's values and preferences and (1 studies of resources utilization: Evidence regarding patient's values and preferences, utilization of resources involving the interventions, and feasibility is mandatory to elaborate the recommendations. This is crucial to successfully adapt an international document to the local .context

Interventions that need rigorous controlled trials include Decision analysis tools, Audit and (2 feedback, Information dissemination, Antenatal education on natural childbirth with training in breathing and relaxation techniques, nurse-led applied relaxation training program, Psychosocial couple-based prevention program, childbirth training workshop, Pelvic floor muscle training exercises, laborist model of obstetric care, psychoeducation (face to face or by telephone), education of public health nurses on childbirth classes, Prenatal education for husbands, cognitive behavioral therapy and childbirth psychotherapy, midwife-led continuity model of care, continuous one-to-one intrapartum support, simulation-based obstetrics and .neonatal emergency training, physical activity-based interventions



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